

Extension of the Bellman Principle to Pareto Fronts in Multi-Objective Path Problems

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1 Notation

Let $G = (V, E)$ be a directed graph, where V is the set of nodes and E the set of edges. The state of the system is given by a node $x \in V$. The actions available at x correspond to outgoing edges e_{xj} , where j indexes the set of neighbors of x . Each edge e_{xj} is assigned two non-negative costs: d_{xj} (the first cost), and p_{xj} (the second cost).

A *path* from node x_0 to node x_n is denoted by $\sigma_{x_0, x_n} = (e_{x_0 x_1}, e_{x_1 x_2}, \dots, e_{x_{n-1} x_n})$. Composition of a path with an edge is written using the symbol $\circ$: for example,

$$\sigma_{x_0, x_{n-1}} \circ e_{x_{n-1}, x_n} = \sigma_{x_0, x_n}.$$

The *cost vector* of a path σ_{x_0, x_n} is defined by

$$C(\sigma_{x_0, x_n}) = \left(\sum_{k=1}^n d_{x_{k-1} x_k}, \sum_{k=1}^n p_{x_{k-1} x_k} \right). \quad (1)$$

2 Pareto Optimality and Value Set

A cost vector $c = (d, p)$ is said to *dominate* another vector $c' = (d', p')$ if $d \leq d'$ and $p \leq p'$ with at least one strict inequality. A path $\sigma_{x_0, x}$ is *Pareto optimal* if there does not exist any other path $\sigma'_{x_0, x}$ from x_0 to x with $C(\sigma'_{x_0, x})$ dominating $C(\sigma_{x_0, x})$.

The **value set** of node x is defined as the set of cost vectors achieved by Pareto-optimal paths from x_0 :

$$V(x) = \{C(\sigma_{x_0, x}) : \sigma_{x_0, x} \text{ is Pareto optimal}\}. \quad (2)$$

3 Bellman Principle for Pareto Fronts

We now state and prove the multi-objective extension of the Bellman principle for the value sets.

Proposition 1 (Bellman Principle for Pareto Fronts). *Let x_n be a node with incoming edges from the set of predecessors x_{n-1} (i.e., $e_{x_{n-1}, x_n} \in E$). Then the value set at x_n satisfies*

$$V(x_n) = \text{ParetoFront} \left(\bigcup_{x_{n-1}: e_{x_{n-1}, x_n} \in E} \{ c + (d_{x_{n-1}, x_n}, p_{x_{n-1}, x_n}) \mid c \in V(x_{n-1}) \} \right), \quad (3)$$

where $c + (d, p)$ denotes componentwise addition, and $\text{ParetoFront}(\cdot)$ denotes the extraction of the set of non-dominated vectors.

Proof. Let us fix the terminal node x_n . By the definition of a path, any σ_{x_0, x_n} must end with some predecessor x_{n-1} and the edge e_{x_{n-1}, x_n} :

$$\sigma_{x_0, x_n} = \sigma_{x_0, x_{n-1}} \circ e_{x_{n-1}, x_n}.$$

By equation (1), the cost vector for such a path can be written as

$$C(\sigma_{x_0, x_n}) = C(\sigma_{x_0, x_{n-1}}) + (d_{x_{n-1}, x_n}, p_{x_{n-1}, x_n}). \quad (4)$$

Therefore, the set of all cost vectors that can reach x_n is the union over all predecessors x_{n-1} and over all possible cost vectors c from Pareto-optimal paths to x_{n-1} , with addition of the last edge's cost:

$$S(x_n) = \bigcup_{x_{n-1}: e_{x_{n-1}, x_n} \in E} \{ c + (d_{x_{n-1}, x_n}, p_{x_{n-1}, x_n}) \mid c \in V(x_{n-1}) \}. \quad (5)$$

The value set $V(x_n)$ must consist of those elements of $S(x_n)$ that are not dominated by any other element in $S(x_n)$, i.e., the Pareto front of $S(x_n)$. Indeed,

- If c is Pareto-optimal at x_{n-1} , then $c + (d_{x_{n-1}, x_n}, p_{x_{n-1}, x_n})$ is a possible cost vector to x_n via a path that is Pareto-optimal up to x_{n-1} .
- If a cost vector can be improved or matched in both components by some other extension from potentially another predecessor, then it is not Pareto-optimal at x_n and is discarded.

Thus, to compute $V(x_n)$, it is sufficient and necessary to:

1. Collect all cost vectors in $S(x_n)$ formed by extending all Pareto-optimal vectors from predecessors,
2. extract the non-dominated subset, i.e., the Pareto front.

This precisely matches equation (3), establishing the recursive structure analogous to the classical Bellman principle. $\square$

Summary: We have generalized the Bellman principle to the setting of multi-objective path problems, showing that the Pareto-optimal value set at each node is recursively constructed by extending the Pareto-optimal sets from all predecessors using the respective edge costs, followed by selection of the Pareto front. This provides a constructive and inductive characterization of optimal paths in multi-objective graph models.